

**Classifying Penguin Species using Morphometric Measurements and
Island Location in the Palmer Archipelago**

AI/ML Software Tools & Platforms

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Question

The main question for this project looks at: To what extent can Adelie, Chinstrap, and Gentoo penguins in the palmer Archipelago be classified using physical measurements, and does adding the island of observation improve classification performance beyond physical measurements alone.

Dataset description

For this project, I used the Palmer Archipelago Antarctica penguin data. This dataset was developed as a modern and approachable alternative to the famous Iris dataset, and it works well for classification, visualization, and data wrangling. A major strength of this dataset is that it comes relatively clean and easy to understand, which allowed the analysis to focus more on preparing features, exploring patterns, and building the classification model.

One weakness is that the dataset is small and not evenly balanced across species. The raw data includes 3444 penguins across three species: 152 Adelie, 68 Chinstrap, and 124 Gentoo penguins. The observations are also distributed across only 3 islands, and some species are strongly associated with specific islands. If island is highly associated with species, the model may partly learn where a penguin was observed rather than relying on the physical measurements.

There were two versions available: a simpler and more comprehensive version. I chose the simpler because it included the variables most relevant to my core question: species, island, culmen (beak) length in mm, culmen depth in mm, flipper length in mm, body mass.

Data wrangling, Cleaning, feature engineering, and split

I began by inspecting the raw data for its shape, column names data types, and missing values recognized by pandas. I also check for any place holder missing values such as "NA", "N/A", "none", "null", ".", blank spaces, or any other common missing value tokens. Then placeholders were converted to pandas missing values so they would be recognized and handled consistently. I selected only columns that were needed for the research question: species, island, culmen length, culmen depth, flipper length, and body mass. Only rows missing values in these modeling columns were then dropped. I also chose not to include the sex column because it introduced additional missing values and was outside

the main focus for this project, classifying species using physical measurements and island.

For feature engineering, the target variable was species, since the model was built to predict Adelie, Chinstrap, or Gentoo penguins. The predictor variables were culmen length, culmen depth, flipper length, body mass, and island. These features were selected because the EDA showed meaningful differences in penguin measurements and island distribution across species. I scaled numeric measurement variables for logistic regression, and island was one-hot encoded due to it being a categorical variable. The cleaned data was split into training and test sets using an 80/20 split. Stratification was also used so the species proportions remained similar in both sets, this was important because chinstrap penguins represented a smaller share of the data than Adiele and Gentoo. This setup focuses on a fair evaluation. The model is only tested on withheld data and has a similar species distribution to the training data.

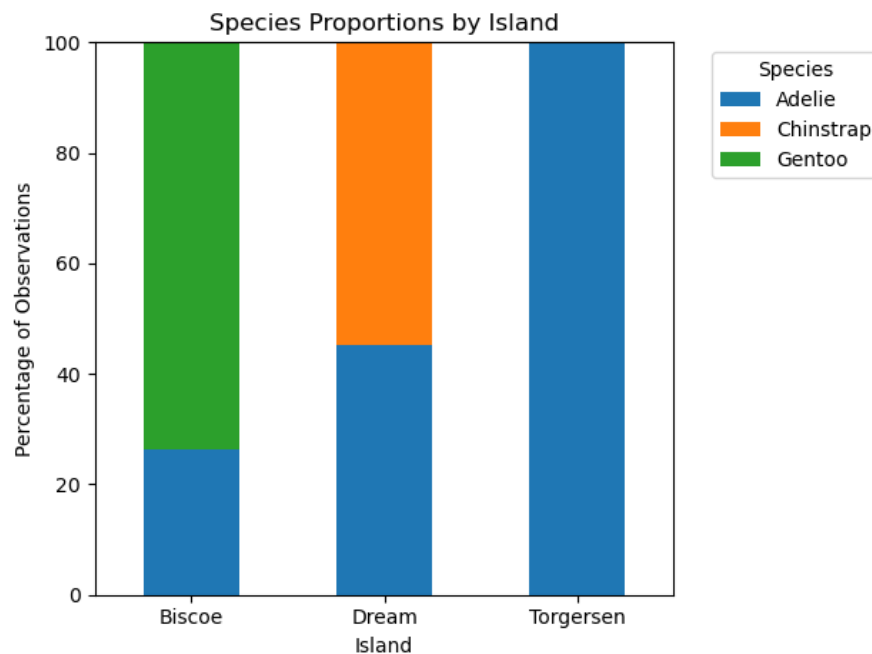
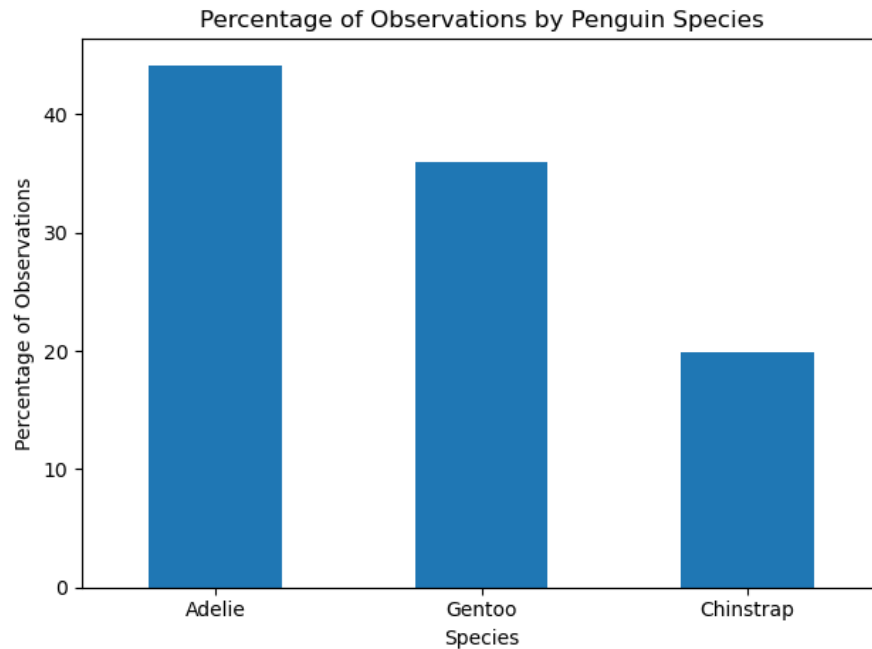
EDA

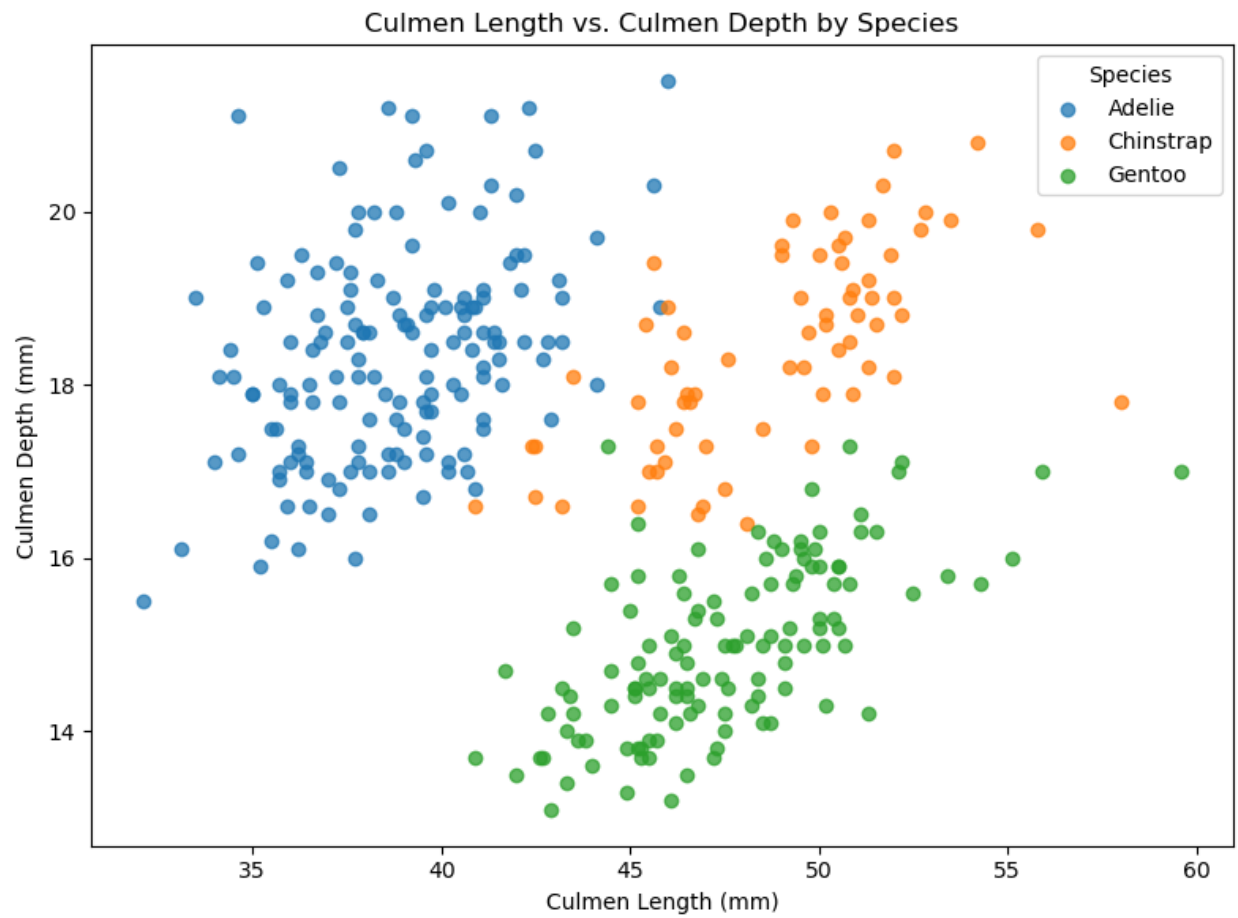
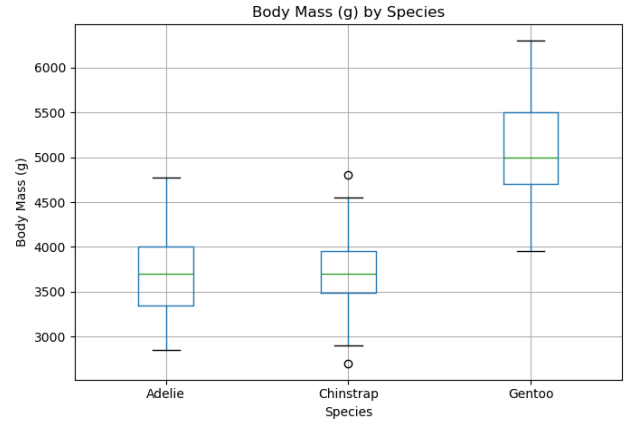
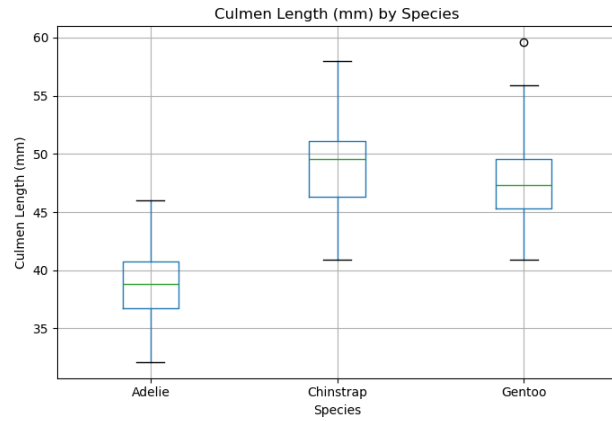
The EDA showed that the cleaned dataset was not evenly distributed across species. Adelie penguins accounted for the largest share with 151 observations, or 44.15% of the cleaned dataset, followed by Gentoo with 123 observations, or 35.96%. Chinstrap penguins had the fewest observations with 68, representing 19.88% of the dataset. This imbalance is important to acknowledge because model performance could be influenced by the lower number of Chinstrap observations.

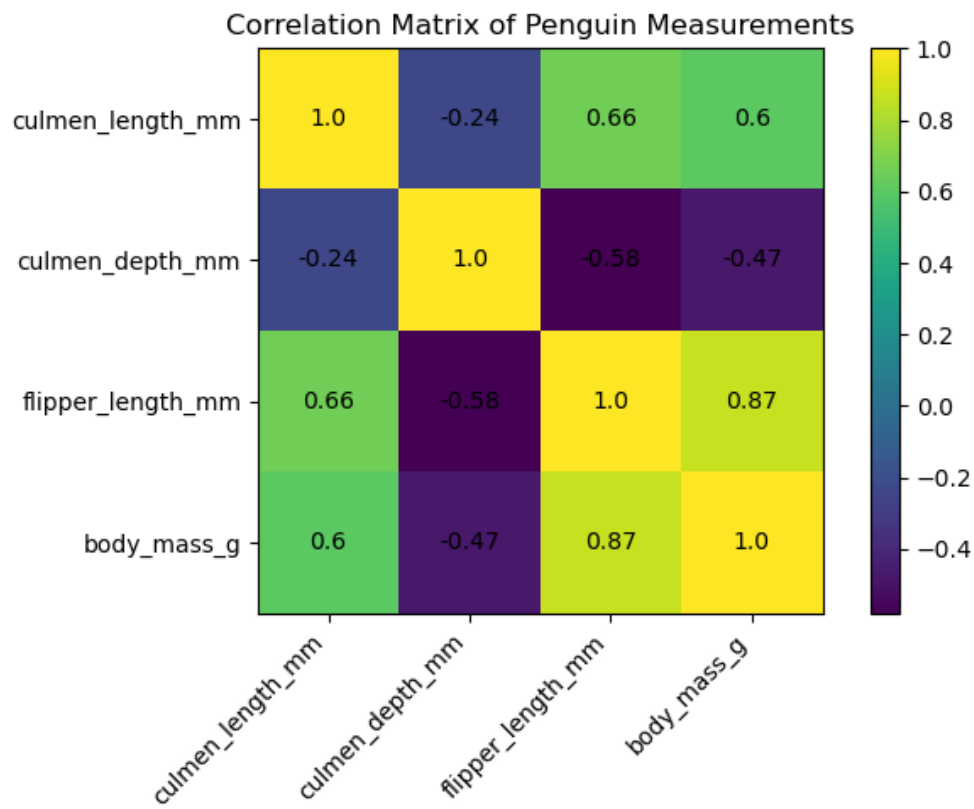
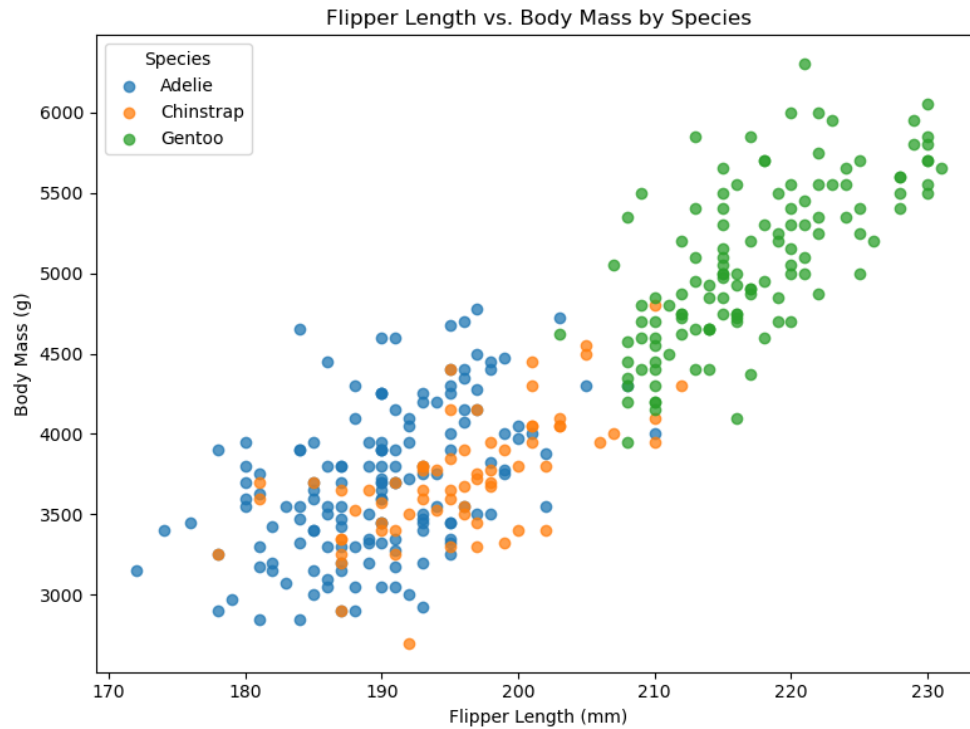
I also explored species proportions by island to understand whether island and species were closely linked. The distribution differed strongly by island. Biscoe contained both Adelie and Gentoo penguins, with Gentoo making up the majority. Dream contained Adelie and Chinstrap penguins, with Chinstrap making up a slight majority. Torgersen contained only Adelie observations. This suggests that island may be useful for classification because some species are strongly associated with specific islands. However, island should be interpreted carefully because it describes where the penguin was observed, not a biological trait.

The measurement summaries and visualizations also showed clear physical differences between species. Gentoo penguins were the most distinct, with the highest average body mass and longest average flipper length. Adelie penguins had the shortest average culmen length, while Chinstrap penguins had the longest average culmen length. Adelie and Chinstrap were harder to distinguish by body mass and flipper length alone, but culmen length and culmen depth helped separate them more clearly.

The scatterplots supported these patterns. Culmen length and culmen depth showed visible clustering by species: Adelie penguins tended to have shorter and deeper culmens, Gentoo penguins tended to have longer and shallower culmens, and Chinstrap penguins tended to have longer and deeper culmens. Flipper length and body mass also showed a positive relationship, with Gentoo forming the clearest separate cluster. Overall, the EDA supported using physical measurements as the main predictors in the classification model and testing island as an additional feature.







Modeling and Prediction

I used a multinomial logistic regression because the target variable, species, has 3 categories: Adelie, Chinstrap, and Gentoo. Two Models were created to address the research question. Model 1 used only physical measurement variables: culmen length, culmen depth, flipper length, and body mass. Model 2 used those same physical measurements but also included island to test whether location improved classification performance.

Because logistic regression requires numeric outputs, Island was one-hot encoded so the model would treat each island as a category rather than a ranked value. The measurement columns were also scaled because they were measured on different values, such as millimeters for culmen and flipper measurements and grams for body mass. Both models were trained on the training set and evaluated using the test set and five-fold cross-validation.

Final Model Interpretation and Conclusion

The purpose of this project was to determine which penguin characteristic help classify the species and how accurately Adelie, Chinstrap, and Gentoo penguins can be predicted using a multinomial logistic regression model. My analysis focused on morphic measurements first, culmen length, culmen depth, flipper length, and body mass. A second model then added island as an additional predictor to test whether location improved classification performance.

Model 1 used only the four physical measurement variables. The multinomial logistic regression performed extremely well, achieving 100% accuracy on the test set by correctly classifying all 69 test observations. It had a mean accuracy of 98.83% and standard deviation of 0.0058 across a five-fold cross-validation check. This suggests that the penguin species in this dataset is highly classifiable using physical measurements alone. The finding was consistent with the EDA, where Gentoo penguins were clearly separated by body mass and flipper length, while Adelie and Chinstrap were more clearly separated by culmen length and depth.

Model 2 used both physical measurements and island. This model produced a slightly higher mean cross-validation accuracy of 99.42% and a standard deviation of 0.0071, an increase of roughly 0.0059 compared with model 1. Island added some predictive value, but the improvement was smaller because physical measurements already classified the species with very high accuracy.

Model Comparison Summary

Model	Mean CV Accuracy	CV Standard Deviation
Model 1: Measurements Only	0.9883	0.0058
Model 2: Measurements + Island	0.9942	0.0071

From a scientific perspective, these results suggest that species-level differences in penguin body structure are strong enough to support an accurate classification within this dataset. Island was also useful, but should be interpreted carefully. Unlike the other predictors, island is not a biological measurement of the penguin. It is a location-based variable that reflects where the observation was recorded. For example, Torgersen only had recorded observations of Adelie penguins, suggesting that some species were strongly associated with specific islands. Island may help prediction, but it does not explain species differences in the same way that the physical characteristics do.

The main limitation of this project is the small and structured nature of the dataset. While the model performed exceptionally well on the cleaned Palmer Penguins dataset, that does not mean it would perform equally well on new data collected. The dataset was not evenly balanced across species, with Chinstrap penguins representing the smallest group. Stratified splitting and cross-validation helped address this, but a smaller number of Chinstrap observations still limits generalization.

Useful next steps would include testing the model on additional penguin observations or comparing its performance against other classification methods, such as decision trees or random forests. Another possible extension would be to include sex as a predictor, since male and female penguins may differ in body size.

This project was completed in Python using pandas for data wrangling, Matplotlib for visualizations, and scikit-learn for splitting, preprocessing, logistic regression, and cross-validation. Working in VS code with interactive Python cells was especially useful because it kept the analysis organized while allowing each step in the EDA and modeling process to be informed by the previous output.

References:

Dataset

Pandey, P. (n.d.). *Palmer Archipelago (Antarctica) penguin data* [Dataset]. Kaggle.
<https://www.kaggle.com/datasets/parulpandey/palmer-archipelago-antarctica-penguin-data/data>

Data originally collected and made available by Dr. Kristen Gorman and the Palmer Station, Antarctica Long Term Ecological Research Network.

AI / Coding Support Tool

OpenAI. (2026). *ChatGPT* [Large language model].
<https://chat.openai.com/>

ChatGPT was used as a support tool for ML-specific coding and debugging, sanity checking code structure and outputs, and developing background understanding of machine learning concepts, modeling approaches.